Lecture ~~8~~ 9: Logistic regression, more classification

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Machine Learning II: 46-927

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Announcements

- Homework is due Wednesday. There will be a quiz on Friday. Everything is back to the usual schedule.

Today

Today we'll pick back up with logistic regression, and then dive further into classification metrics.

- Logistic regression
- Classification:
  - Imbalanced data
  - Classification metrics

Where we were

$\{1, 2, \dots K\}$

We switched from regression to classification.

Now we observe pairs $(X, Y) \sim \mathcal{P}$, with $X \in \mathbb{R}^p$ and $Y \in \{0, 1\}$ (or sometimes $\{-1, 1\}$).

Based on n observed pairs, we estimate a function $\hat{f}$ that "guesses Y well" for future $(X, Y) \sim \mathcal{P}$.

K-class

In a classification problem, we can make $K(K-1)$ different kinds of mistakes! This leads to two different $K \times K$ matrices reflecting classification performance.

Confusion matrix and loss

Confusion Matrix, which tabulates counts of what actually happened.

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	True Positive	False Positive
Guess $Y = 0$	False Negative	True Negative

2×2

Loss Matrix, which tabulates how bad each outcome combination is for you.

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	0	L_{01}
Guess $Y = 0$	L_{10}	0

2×2

Confusion matrix: error rates

	True $Y = 1$	True $Y = 0$
Guess $Y = 1$	True Positive	False Positive
Guess $Y = 0$	False Negative	True Negative

We are interested in many different quantities in the confusion matrix.
The most famous is the misclassification rate: $(FN + FP)/n$

If False positives and False negatives are equally bad, then minimizing the misclassification rate is a good goal.

If we set our loss to have $L_{01} = L_{10} = 1$ and $L_{00} = L_{11} = 0$, minimizing the expected loss will be the same as minimizing the expected misclassification rate.

Statistical decision theory

0-1 loss

We showed that the expected misclassification error (or expected prediction error),

$$\mathrm{EPE} = \mathbb{E}\left(\mathcal{L}(Y, \hat{f}(X))\right) = \mathbb{E}\left(\mathbf{1}_{Y \neq f(X)}\right),$$

is minimized by the ideal rule:

$$f(x) = \underset{g \in \{1,\ldots,K\}}{\operatorname{argmax}} \; \mathbb{P}(Y = g | X = x),$$

if we knew $\mathbb{P}(Y = g | X = x)$ for all g. This is called the Bayes Classifier, and the error rate it achieves is the Bayes Rate.

Building a classifier

Combining this with Bayes' theorem leads to two formulations of an ideal classifier:

2 classes threshold at 1/2

$$\hat{f}(x) = \boxed{\underset{j=1,\ldots K}{\operatorname{argmax}}\ \mathrm{P}(Y = j | X = x)}$$

$$= \underset{j=1,\ldots K}{\operatorname{argmax}}\ \mathrm{P}(X = x | Y = j) \cdot \pi_j$$

1. **Discriminative models:** We can try to estimate $\mathbb{P}(Y = j | X = x)$ directly. (Discriminative models)

2. **Generative models:** We could try to estimate the distributions $\mathbb{P}(X = x | Y = j)$ for each class.

Logistic regression

$E(Y|X=x)$

How should we think about modeling $\boxed{\mathbb{P}(Y=k|X=x)}$ directly?

If we only had a few x values, we could directly look at Y conditional on each one:

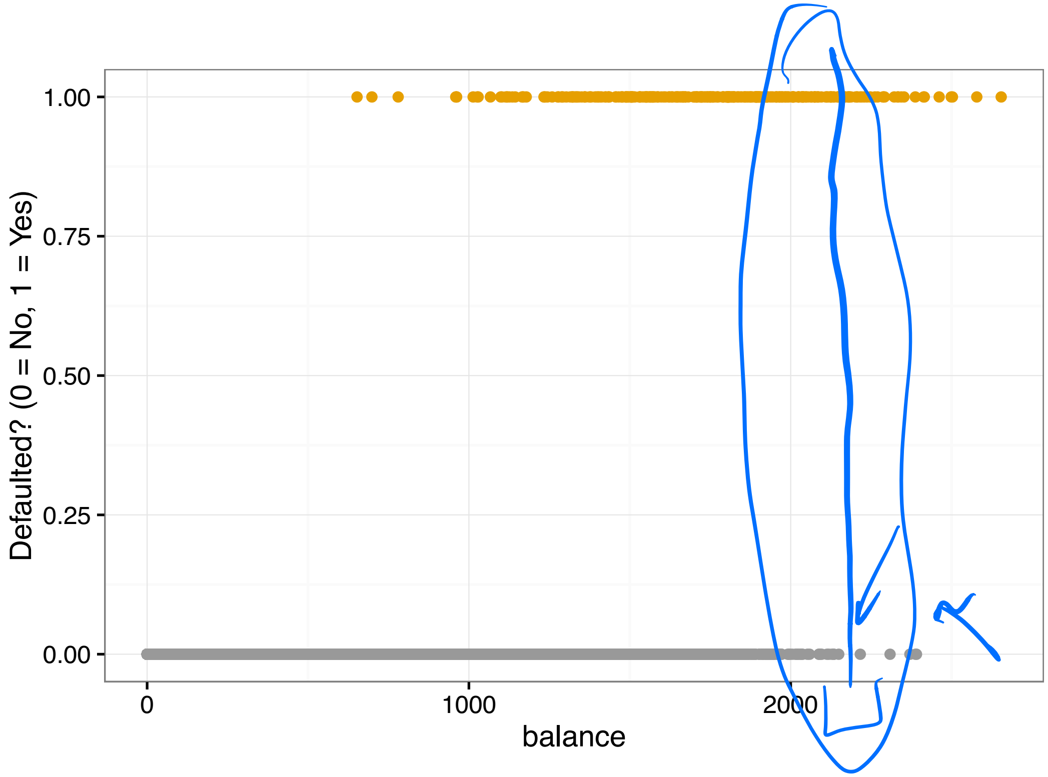
Y

X	Default=No	Default=Yes
Student=No	6850	206
Student=Yes	2817	127

$$P(\text{Default}|\text{Student}) = \boxed{\frac{127}{127+2817}}$$

$$P(\text{Default}|\text{Not Student}) = \frac{206}{206+6850}$$

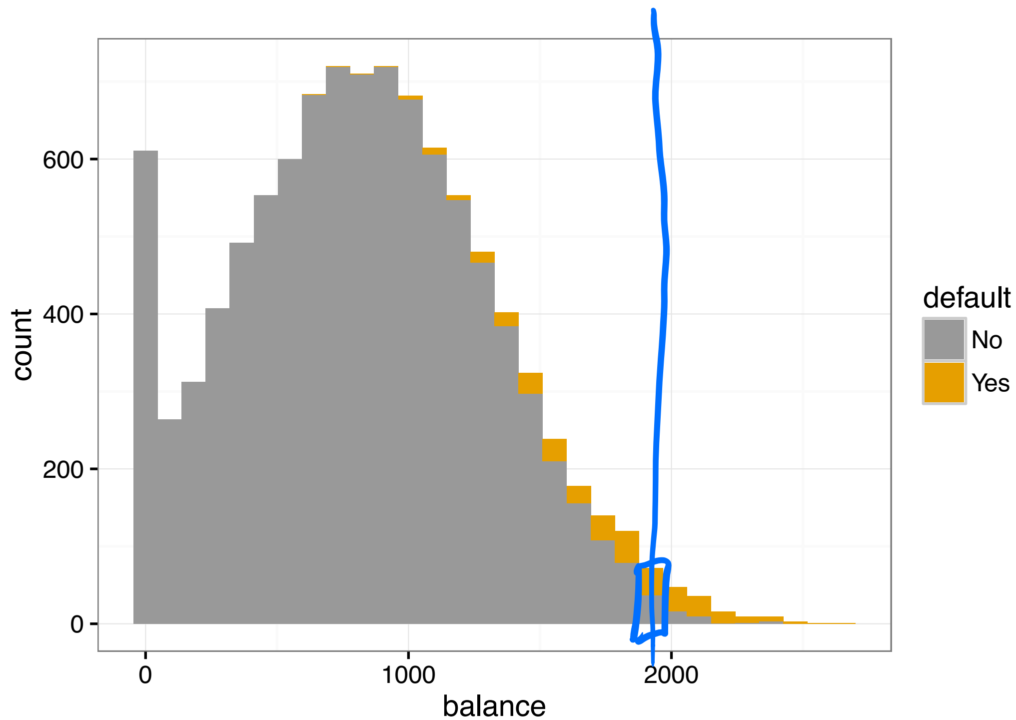
Discriminative models



If we have many values of x, we can't directly look at $P(Y = k|X = x)$ for each x. We need some way to pool information from *similar* points.

This should remind you of regression.

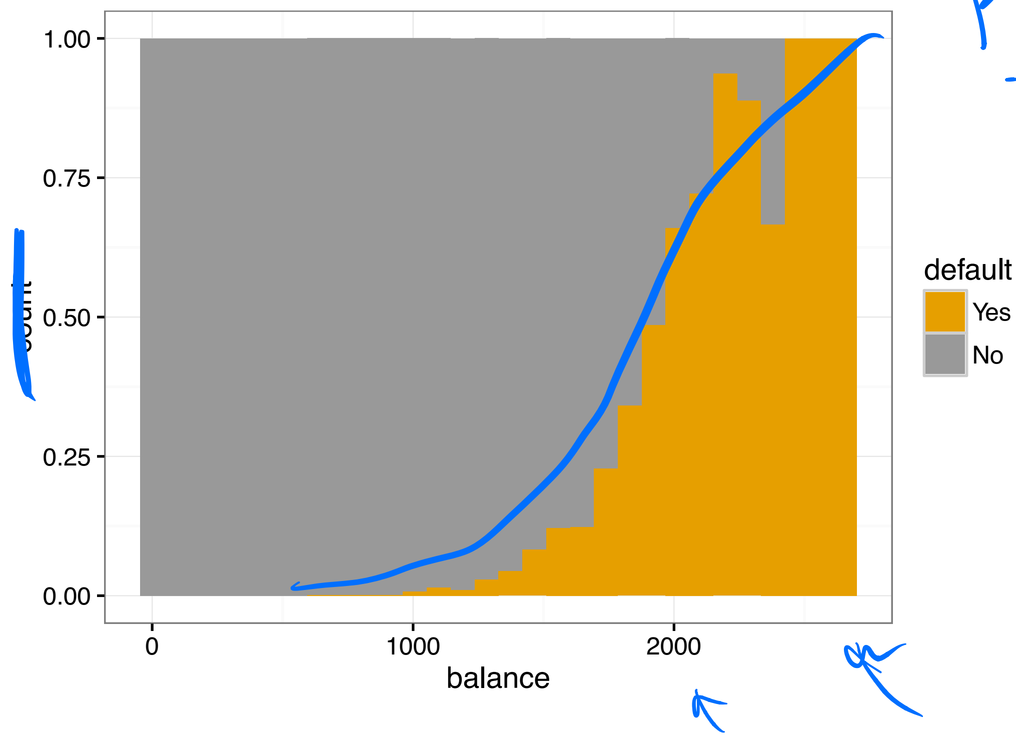
How should we model $P(Y = 1|X = x)$ for a continuous X?



A histogram is not quite the right picture for what happens conditional on X, but we can start to see that most individuals do not default, and that large balance seems to be related to default.

This is a conditional density plot. We look at the *probability of default* within each bin.

$P(Y=1|X=x) = \hat{f}(x)$

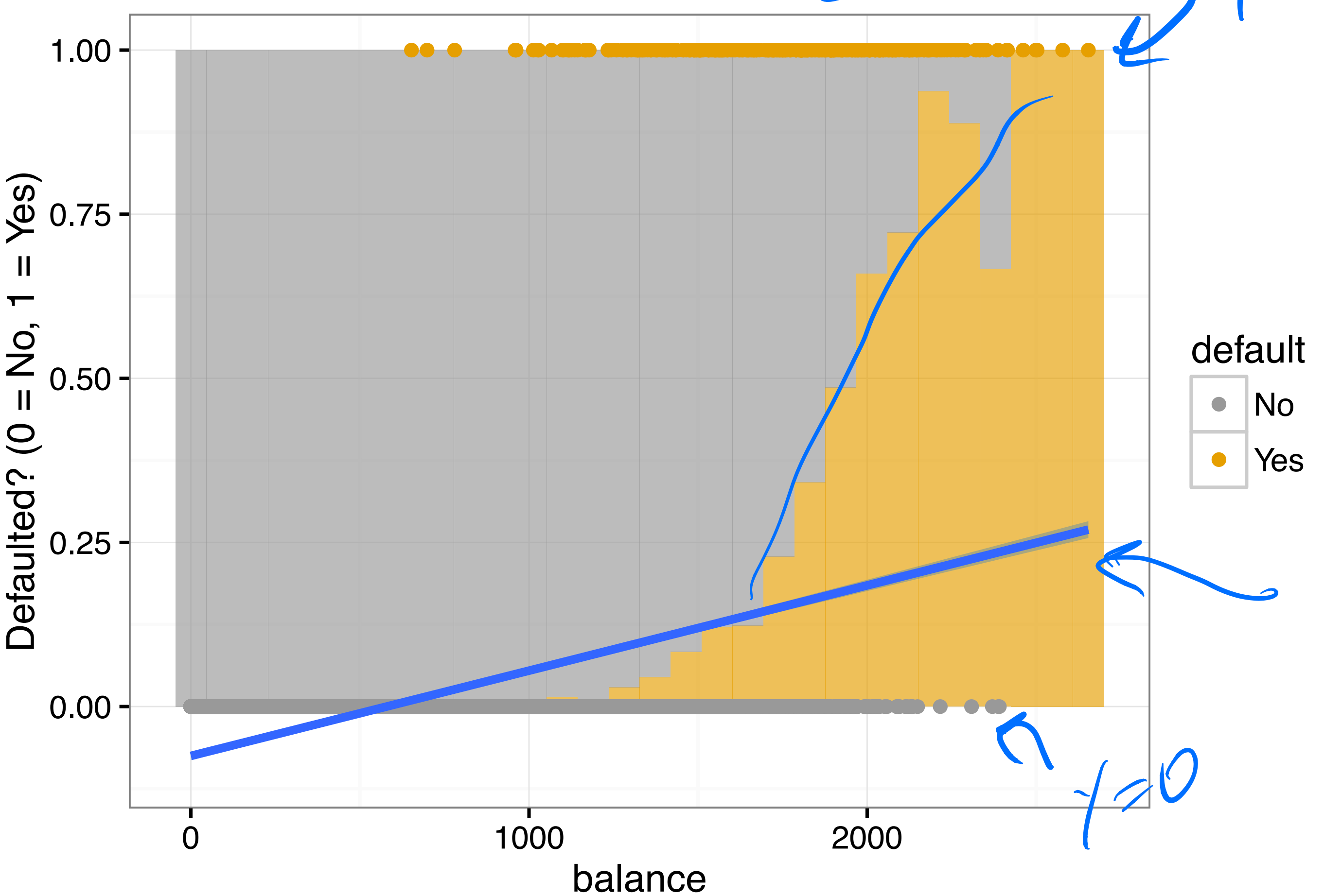


This is a (binned) plot of $P(Y = 1|X = x)$! This is clearly a natural way to think about classification. If I know which bin you are in (X), I can look at how likely you are to default.

How should I build a model of this?

We could try a linear model:

$$P(Y = 1|x = x) = \beta_0 + \beta_1 x$$

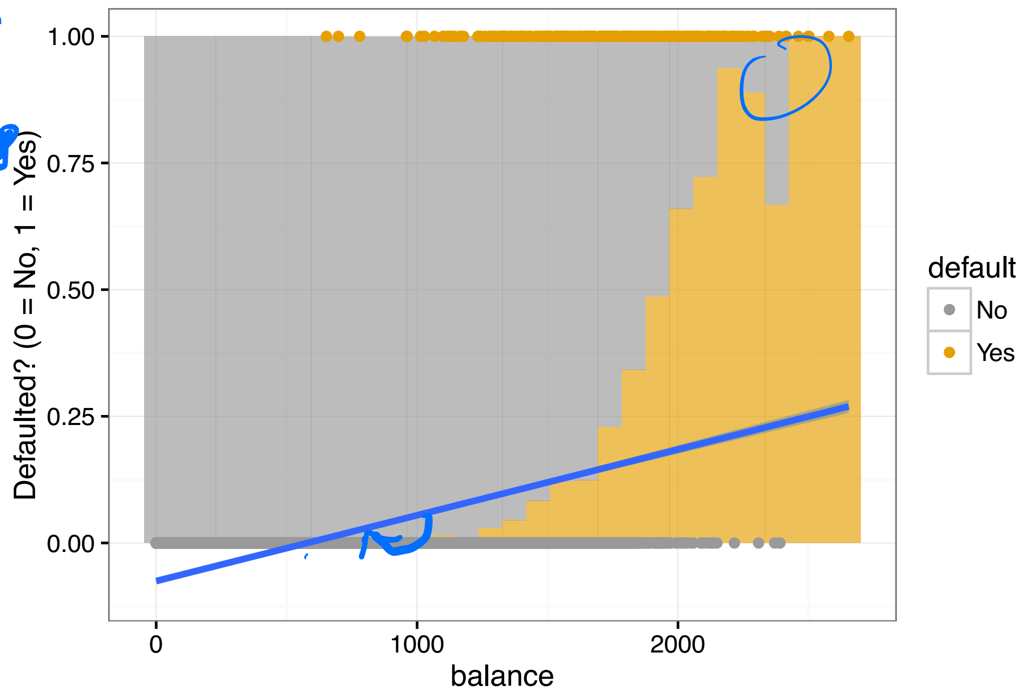


OLS

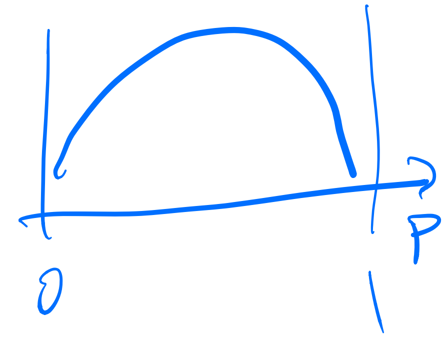
w_i

$\min \sum_i (y_i - x_i^T \beta)^2$

$\{0,1\}$



$\frac{p(1-p)}{n}$



Are you happy?

- Predictions outside $[0, 1]$ don't really make sense.
- Interpretations of β are strange.
- Least squares doesn't really make sense for estimation.

Linear regression doesn't work very well for estimating $P(Y = 1|X = x)$, since

- It doesn't make sense to extrapolate outside of $[0, 1]$.
- Least squares is an odd way to approximate probabilities.

We still like the idea of forming linear functions of our data, $\beta_0 + x_1\beta_1$ (who doesn't?).

We want a way to squish that linear function back into $[0, 1]$:

$$P(Y = 1|X = x) = \text{squish}(\beta_0 + \beta_1 x_1)$$

$\mathbb{R} \to [0,1]$

monotone.

Logistic regression

$p = P(Y=1|X=x)$

In logistic regression, we model

$$\log\left\{\frac{P(Y=1|X=x)}{P(Y=0|X=x)}\right\} = \beta_0 + \beta_1 x + \cdots = \beta^T x$$

for some unknown $\beta \in \mathbb{R}^p$, which we will estimate directly

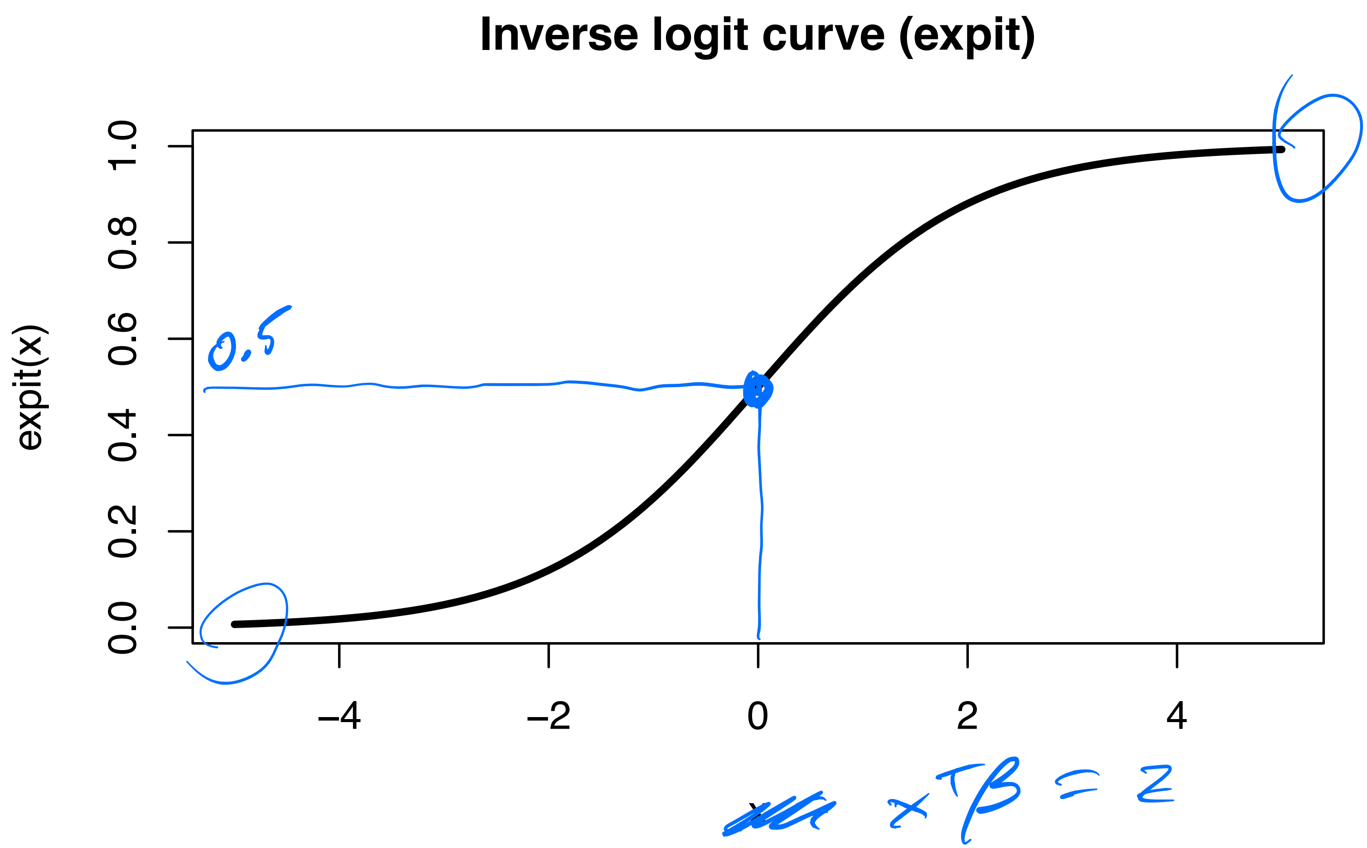
Note that $P(Y=0|X=x) = 1 - P(Y=1|X=x)$, and

$$\underbrace{\log\left(\frac{p}{1-p}\right)}_{\text{logit}} = \beta^T x \quad \Leftrightarrow \quad \frac{p}{1-p} = e^{\beta^T x} \Rightarrow p = \frac{e^{\beta^T x}}{1+e^{\beta^T x}}$$

$$\Leftrightarrow \quad p = \frac{1}{1+e^{-\beta^T x}}$$

So our model is equivalent to

$$P(Y=1|X=x) = \frac{\exp(\beta^T x)}{1+\exp(\beta^T x)} \in [0,1]$$



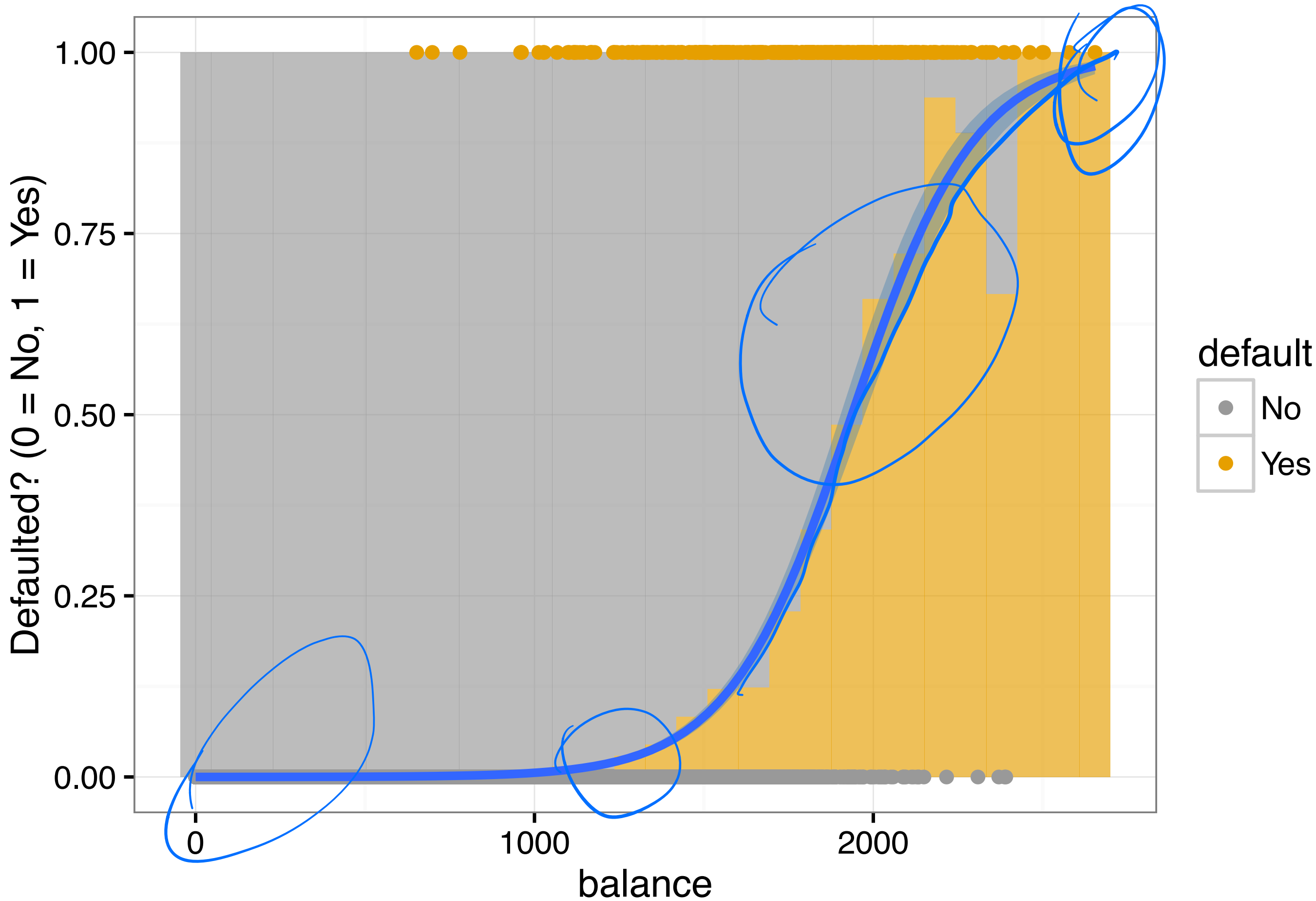
The function

$$\text{logit}^{-1}(z) = \text{expit}(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}$$

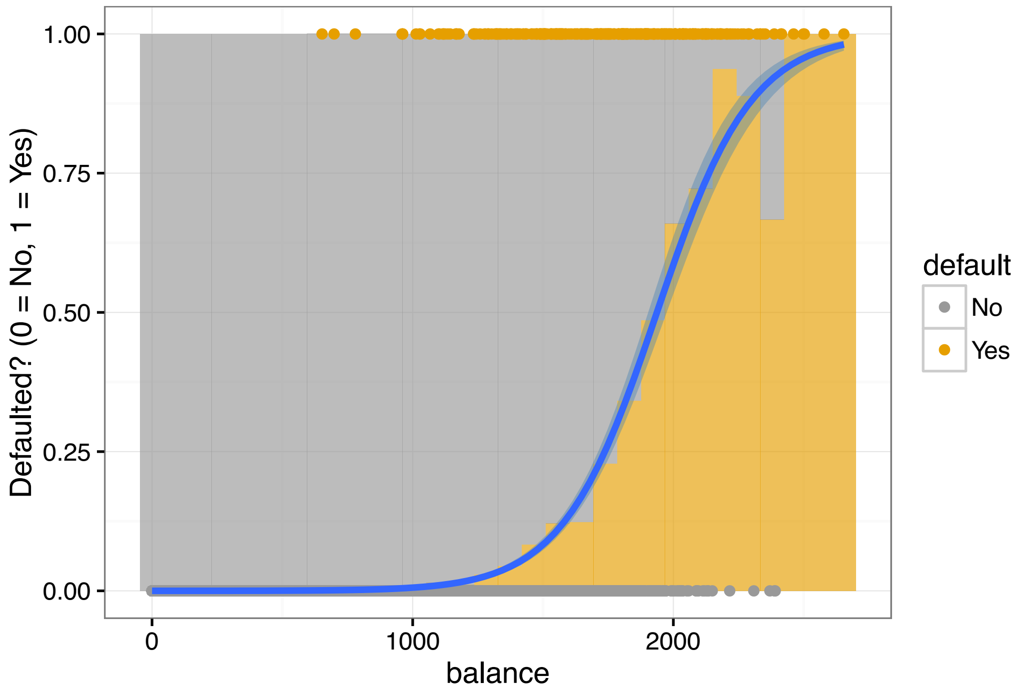
is our desired "squishing" function, transforming real numbers into $[0,1]$.

Logistic regression and the Default data

The logistic fit gives a much more reasonable estimate of $P(Y = 1|X = x)$!



Logistic regression and the Default data



$$\log \frac{P(\text{Default}|\text{Balance})}{1 - P(\text{Default}|\text{Balance})} = \beta_0 + \beta_1 \cdot \text{Balance}$$

Logistic regression: Estimated probabilities

Once we have estimated $\hat{\beta}_0, \hat{\beta}_1$, we can estimate conditonal probabilities:

$$\mathrm{P}(Y = 1 | X = x) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 x)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 x)}$$

For a balance of \$1000 or \$2000,

$$\hat{p}(1000) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1000)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 1000)} = 0.00576$$

$Y = 0$

$$\hat{p}(2000) = \frac{\exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 2000)}{1 + \exp(\hat{\beta}_0 + \hat{\beta}_1 \cdot 2000)} = 0.586$$

$Y = 1$

Interpretation of logistic regression coefficients

We start to see that the coefficients are *interpretable*.

$\log\left(\frac{p}{1-p}\right)$ increasing.

We have modeled

$$\log \frac{P(Y=1|X=x)}{P(Y=0|X=x)} = \beta_0 + \beta_1 x_1$$

$\frac{p}{1-p}$ odds

The left side, $\log \frac{P(Y=1|X=x)}{P(Y=0|X=x)}$, is called the *log-odds* that $Y=1$.

This means that the odds that $Y=1$,

$$\frac{P(Y=1|X=x)}{P(Y=0|X=x)} = e^{\beta_0+\beta_1 x_1} = e^{\beta_0} e^{\beta_1 x_1}$$

Increasing x_1 by one unit increases the estimated odds that $Y=1$ by e^{β_1}.

Classification by logistic regression

Suppose that we fit a logistic regression, estimating $\hat{\beta}$. How do we classify?

Recall that our optimal classifier chooses

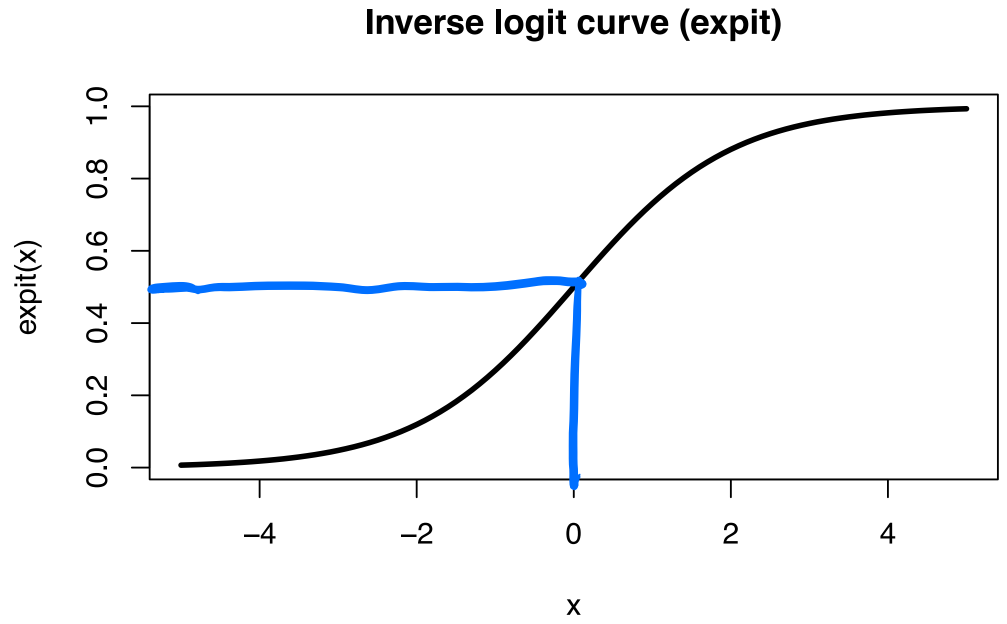
$$\underset{k}{\operatorname{argmax}}\, P(Y = k|X = x)$$

to minimize 0-1 loss.

Logistic regression gives us an estimate of $P(Y = k|X = x)$! We can just pick the category with the biggest value.

$$\hat{f}(x) = \begin{cases} 1 & \text{if } \frac{\exp(x^T\hat{\beta})}{1+\exp(x^T\hat{\beta})} > 0.5 \\ 0 & \text{otherwise} \end{cases}$$

Classification by logistic regression



Remember that logit and expit are monotonically increasing! This gives us a much simpler rule!

$$\underbrace{\frac{\exp(x^T\hat{\beta})}{1+\exp(x^T\hat{\beta})}}\ > 0.5 \quad \Leftrightarrow \quad \overbrace{x^T\beta > 0}$$

Classification by logistic regression

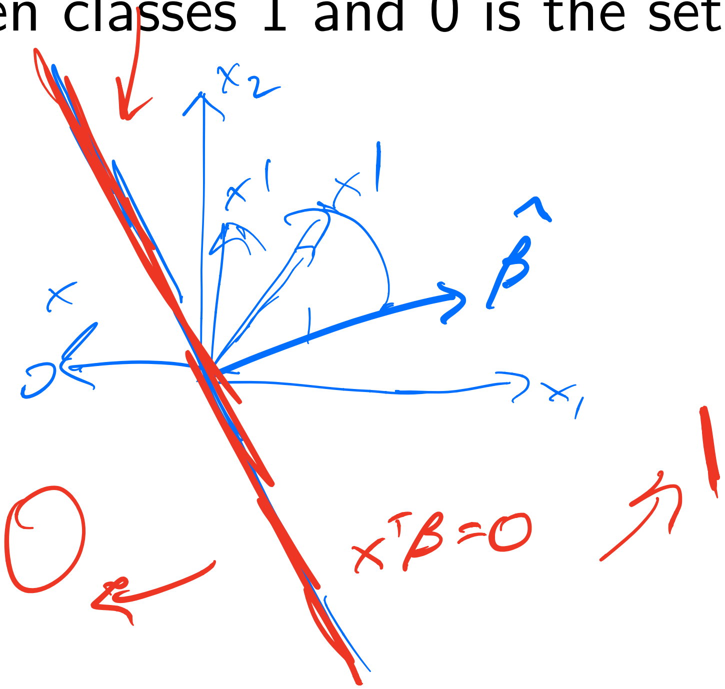
$x^T\beta = \|x\|\,\|\beta\| \cos\theta$

This gives our final decision rule

$$\hat{f}(x) = \begin{cases} 1 & \text{if } x^T\hat{\beta} > 0 \\ 0 & \text{otherwise} \end{cases}$$

Therefore the decision boundary between classes 1 and 0 is the set of all $x \in \mathbb{R}^p$ such that

This is a point in $\mathbb{R}^1$ or a line in $\mathbb{R}^2$.



Estimating logistic regression coefficients

To actually estimate the $\hat{\beta}$, we just use maximum likelihood!

Suppose that we are given an i.i.d. sample (x_i, y_i), $i = 1, \ldots n$. Here y_i denotes the class $\in \{0, 1\}$ of the ith observation. Then

$$\mathcal{L}(\beta) = \prod_{i=1}^{n} \mathrm{P}(C = y_i | X = x_i)$$

the likelihood of these n observations, so the log likelihood is

$$\ell(\beta) = \sum_{i=1}^{n} \log \mathrm{P}(C = y_i | X = x_i)$$

We just plug in our logistic model for these probabilities and optimize.

$$p(x_i) = \frac{e^{x_i^T\beta}}{1+e^{x_i^T\beta}}$$

The log likelihood can be written as

$$\ell(\beta) = \sum_{i=1}^{n} \log \mathrm{P}(C = y_i | X = x_i) = \sum_{i=1}^{n} \log\left[p(x_i)^{y_i}\left(1-p(x_i)\right)^{1-y_i}\right]$$

$$=$$

$$=$$

$$= \underbrace{\sum_{i=1}^{n} \left\{ y_i \cdot (\beta^T x_i) - \log\left(1 + \exp(\beta^T x_i)\right) \right\}}$$

The coefficients are estimated by maximizing the likelihood,

$$\hat{\beta} = \underset{\cancel{\beta\in\mathbb{R}},\, \beta\in\mathbb{R}^p}{\operatorname{argmax}} \sum_{i=1}^{n} \left\{ y_i \cdot (\beta^T x_i) - \log\left(1 + \exp(\beta^T x_i)\right) \right\} - \lambda \|\beta\|_1$$

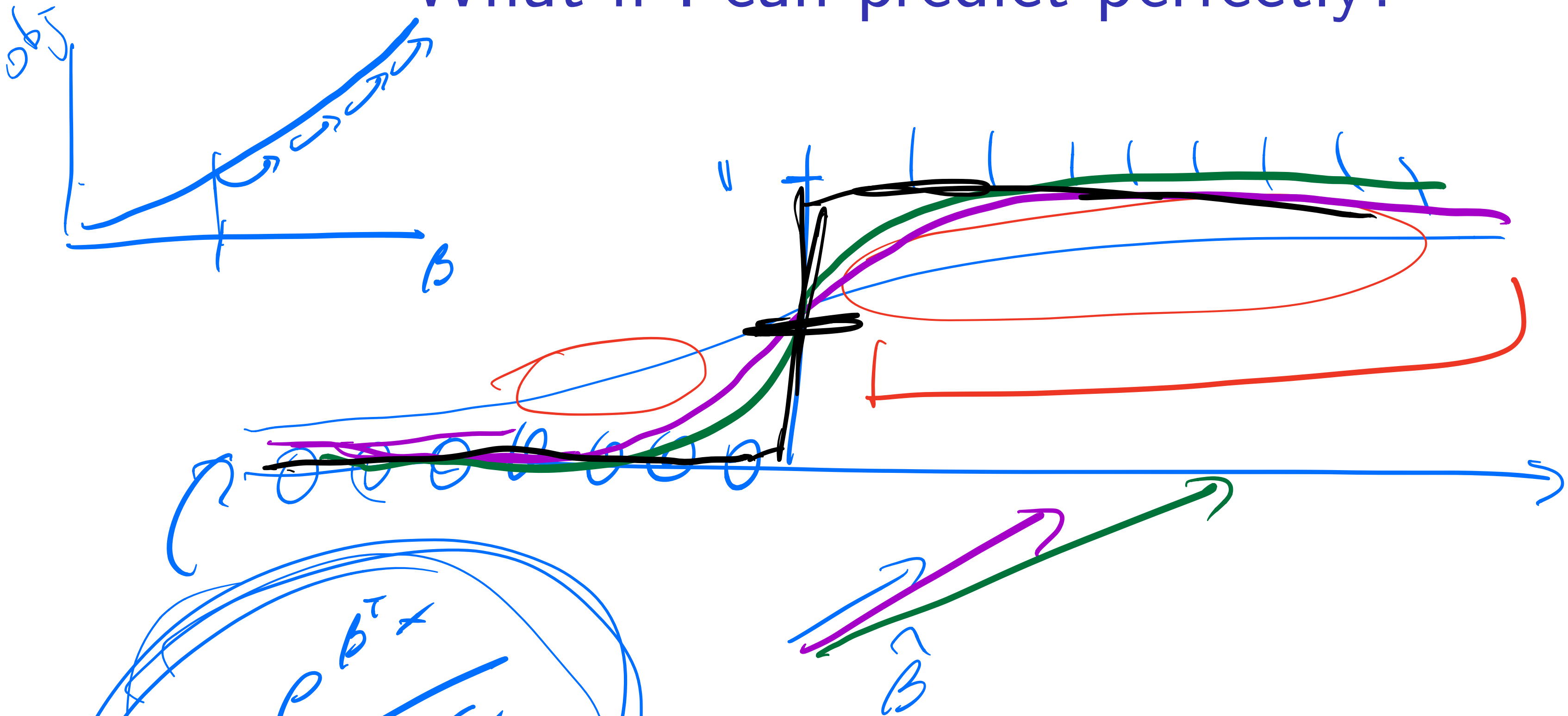
$$\sum_{j=1}^{p} \hat{f}_j(x_{ij})$$

$$\int f''(x)^2 \ldots$$

Iteratively reweighted least squares

$$\sum w_i (y_i - x_i^T\beta)^2$$

What if I can predict perfectly?



x

$\alpha \uparrow$

$$\frac{e^{\beta^T x}}{1+e^{\beta^T x}} = P(Y=1 \mid X=x)$$

What if I have many variables?

Suppose that the number of variables is large. We would expect the variance of our estimated $\hat{\beta}$ to grow!

What could we do to reduce the variance?

regularize! ridge? lasso?

What if we also wanted to pick a sparse, interpretable linear model?

lasso

What if we need a more flexible model?

What if we don't really believe that the log odds increase linearly in each of our variables? What if we don't know how to transform our variables?